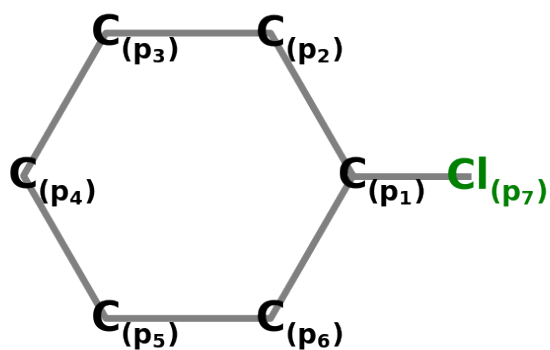


Chlorobenzene



Point Group: **C_{2v}**

p orbitals

Symmetry behavior of the p-orbitals:

$$\Gamma_{red} = 7E \oplus -3C_2 \oplus -7\sigma_v(xz) \oplus 3\sigma_v(yz)$$

$$\Gamma_{irred} = 2A_2 \oplus 5B_2$$

- SALC of B2: p_1
- SALC of A2: $\frac{\sqrt{2}(p_2 - p_6)}{2}$
- SALC of B2: $\frac{\sqrt{2}(p_2 + p_6)}{2}$
- SALC of A2: $\frac{\sqrt{2}(p_3 - p_5)}{2}$
- SALC of B2: $\frac{\sqrt{2}(p_3 + p_5)}{2}$
- SALC of B2: p_4
- SALC of B2: p_7

Put together these SALCs form the following Hamilton matrices:

$$H_{B2} = \begin{bmatrix} \tilde{\alpha} & \sqrt{2}\tilde{\beta} & 0 & 0 & \tilde{\beta}_{Cl} \\ \sqrt{2}\tilde{\beta} & \tilde{\alpha} & \tilde{\beta} & 0 & 0 \\ 0 & \tilde{\beta} & \tilde{\alpha} & \sqrt{2}\tilde{\beta} & 0 \\ 0 & 0 & \sqrt{2}\tilde{\beta} & \tilde{\alpha} & 0 \\ \tilde{\beta}_{Cl} & 0 & 0 & 0 & \tilde{\alpha}_{Cl} \end{bmatrix}$$

$$H_{A2} = \begin{bmatrix} \tilde{\alpha} & \tilde{\beta} \\ \tilde{\beta} & \tilde{\alpha} \end{bmatrix}$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of A2 with energy = $\tilde{\alpha} + \tilde{\beta}$
(

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \text{1. SALC in A2} \\ \text{2. SALC in A2} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{2}p_2}{2} - \frac{\sqrt{2}p_6}{2} \\ \frac{\sqrt{2}p_3}{2} - \frac{\sqrt{2}p_5}{2} \end{bmatrix}$$

$$= \frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} - \frac{p_6}{2}$$

)

- orbital of A2 with energy = $\tilde{\alpha} - \tilde{\beta}$
(

$$\begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \text{1. SALC in A2} \\ \text{2. SALC in A2} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{2}p_2}{2} - \frac{\sqrt{2}p_6}{2} \\ \frac{\sqrt{2}p_3}{2} - \frac{\sqrt{2}p_5}{2} \end{bmatrix}$$

$$= -\frac{p_2}{2} + \frac{p_3}{2} - \frac{p_5}{2} + \frac{p_6}{2}$$

)

- orbital of B2 with energy = $E0(B2)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in B2} \\ 2. \text{ SALC in B2} \\ 3. \text{ SALC in B2} \\ 4. \text{ SALC in B2} \\ 5. \text{ SALC in B2} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{p}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_2}{2} + \frac{\sqrt{2}\mathbf{p}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_3}{2} + \frac{\sqrt{2}\mathbf{p}_5}{2} \right) + ?_4 Norm^2 \mathbf{p}_4 + ?_5 Norm^2 \mathbf{p}_7 \\
 &)
 \end{aligned}$$

- orbital of B2 with energy = $E1(B2)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in B2} \\ 2. \text{ SALC in B2} \\ 3. \text{ SALC in B2} \\ 4. \text{ SALC in B2} \\ 5. \text{ SALC in B2} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{p}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_2}{2} + \frac{\sqrt{2}\mathbf{p}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_3}{2} + \frac{\sqrt{2}\mathbf{p}_5}{2} \right) + ?_4 Norm^2 \mathbf{p}_4 + ?_5 Norm^2 \mathbf{p}_7 \\
 &)
 \end{aligned}$$

- orbital of B2 with energy = $E2(B2)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in B2} \\ 2. \text{ SALC in B2} \\ 3. \text{ SALC in B2} \\ 4. \text{ SALC in B2} \\ 5. \text{ SALC in B2} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{p}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_2}{2} + \frac{\sqrt{2}\mathbf{p}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_3}{2} + \frac{\sqrt{2}\mathbf{p}_5}{2} \right) + ?_4 Norm^2 \mathbf{p}_4 + ?_5 Norm^2 \mathbf{p}_7 \\
 &)
 \end{aligned}$$

- orbital of B2 with energy = $E3(B2)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in B2} \\ 2. \text{ SALC in B2} \\ 3. \text{ SALC in B2} \\ 4. \text{ SALC in B2} \\ 5. \text{ SALC in B2} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{p}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_2}{2} + \frac{\sqrt{2}\mathbf{p}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{p}_3}{2} + \frac{\sqrt{2}\mathbf{p}_5}{2} \right) + ?_4 Norm^2 \mathbf{p}_4 + ?_5 Norm^2 \mathbf{p}_7 \\
 &)
 \end{aligned}$$

- orbital of B2 with energy = $E_4(B_2)$

(

$$[\varphi_1 \text{Norm}^2 \quad \varphi_2 \text{Norm}^2 \quad \varphi_3 \text{Norm}^2 \quad \varphi_4 \text{Norm}^2 \quad \varphi_5 \text{Norm}^2] * \begin{bmatrix} 1. \text{ SALC in B2} \\ 2. \text{ SALC in B2} \\ 3. \text{ SALC in B2} \\ 4. \text{ SALC in B2} \\ 5. \text{ SALC in B2} \end{bmatrix} = [\varphi_1 \text{Norm}^2 \quad \varphi_2 \text{Norm}^2 \quad \varphi_3 \text{Norm}^2 \quad \varphi_4 \text{Norm}^2 \quad \varphi_5 \text{Norm}^2]$$

$$= \varphi_1 \text{Norm}^2 \mathbf{p}_1 + \varphi_2 \text{Norm}^2 \left(\frac{\sqrt{2} \mathbf{p}_2}{2} + \frac{\sqrt{2} \mathbf{p}_6}{2} \right) + \varphi_3 \text{Norm}^2 \left(\frac{\sqrt{2} \mathbf{p}_3}{2} + \frac{\sqrt{2} \mathbf{p}_5}{2} \right) + \varphi_4 \text{Norm}^2 \mathbf{p}_4 + \varphi_5 \text{Norm}^2 \mathbf{p}_7$$

)

s orbitals

Symmetry behavior of the s-orbitals:

$$\Gamma_{red} = 7E \oplus 3C2 \oplus 7\sigma v(xz) \oplus 3\sigma v(yz)$$

$$\Gamma_{irred} = 5A1 \oplus 2B1$$

- SALC of A1: s_1
- SALC of A1: $\frac{\sqrt{2}(s_2+s_6)}{2}$
- SALC of B1: $\frac{\sqrt{2}(s_2-s_6)}{2}$
- SALC of A1: $\frac{\sqrt{2}(s_3+s_5)}{2}$
- SALC of B1: $\frac{\sqrt{2}(s_3-s_5)}{2}$
- SALC of A1: s_4
- SALC of A1: s_7

Put together these SALCs form the following Hamilton matrices:

$$H_{A1} = \begin{bmatrix} \begin{bmatrix} \tilde{\alpha}_s & \sqrt{2}\tilde{\beta}_s & 0 & 0 & \beta_{s_{Cl}} \\ \sqrt{2}\tilde{\beta}_s & \tilde{\alpha}_s & \tilde{\beta}_s & 0 & 0 \\ 0 & \tilde{\beta}_s & \tilde{\alpha}_s & \sqrt{2}\tilde{\beta}_s & 0 \\ 0 & 0 & \sqrt{2}\tilde{\beta}_s & \tilde{\alpha}_s & 0 \\ \beta_{s_{Cl}} & 0 & 0 & 0 & \alpha_{s_{Cl}} \end{bmatrix} \end{bmatrix}$$

$$H_{B1} = \begin{bmatrix} \tilde{\alpha}_s & \tilde{\beta}_s \\ \tilde{\beta}_s & \tilde{\alpha}_s \end{bmatrix}$$

Solving Hückels secular equations, that follow from these H, leads to:

- orbital of B1 with energy = $\tilde{\alpha}_s + \tilde{\beta}_s$

(

$$\begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B1} \\ \text{2. SALC in B1} \end{bmatrix} = \begin{bmatrix} \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{2}s_2}{2} - \frac{\sqrt{2}s_6}{2} \\ \frac{\sqrt{2}s_3}{2} - \frac{\sqrt{2}s_5}{2} \end{bmatrix}$$

$$= \frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} - \frac{s_6}{2}$$

)

- orbital of B1 with energy = $\tilde{\alpha}_s - \tilde{\beta}_s$

(

$$\begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \text{1. SALC in B1} \\ \text{2. SALC in B1} \end{bmatrix} = \begin{bmatrix} -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \end{bmatrix} * \begin{bmatrix} \frac{\sqrt{2}s_2}{2} - \frac{\sqrt{2}s_6}{2} \\ \frac{\sqrt{2}s_3}{2} - \frac{\sqrt{2}s_5}{2} \end{bmatrix}$$

$$= -\frac{s_2}{2} + \frac{s_3}{2} - \frac{s_5}{2} + \frac{s_6}{2}$$

)

- orbital of A1 with energy = $E0(A1)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in A1} \\ 2. \text{ SALC in A1} \\ 3. \text{ SALC in A1} \\ 4. \text{ SALC in A1} \\ 5. \text{ SALC in A1} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{s}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_2}{2} + \frac{\sqrt{2}\mathbf{s}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_3}{2} + \frac{\sqrt{2}\mathbf{s}_5}{2} \right) + ?_4 Norm^2 \mathbf{s}_4 + ?_5 Norm^2 \mathbf{s}_7 \\
 &)
 \end{aligned}$$

- orbital of A1 with energy = $E1(A1)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in A1} \\ 2. \text{ SALC in A1} \\ 3. \text{ SALC in A1} \\ 4. \text{ SALC in A1} \\ 5. \text{ SALC in A1} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{s}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_2}{2} + \frac{\sqrt{2}\mathbf{s}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_3}{2} + \frac{\sqrt{2}\mathbf{s}_5}{2} \right) + ?_4 Norm^2 \mathbf{s}_4 + ?_5 Norm^2 \mathbf{s}_7 \\
 &)
 \end{aligned}$$

- orbital of A1 with energy = $E2(A1)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in A1} \\ 2. \text{ SALC in A1} \\ 3. \text{ SALC in A1} \\ 4. \text{ SALC in A1} \\ 5. \text{ SALC in A1} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{s}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_2}{2} + \frac{\sqrt{2}\mathbf{s}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_3}{2} + \frac{\sqrt{2}\mathbf{s}_5}{2} \right) + ?_4 Norm^2 \mathbf{s}_4 + ?_5 Norm^2 \mathbf{s}_7 \\
 &)
 \end{aligned}$$

- orbital of A1 with energy = $E3(A1)$
(

$$\begin{aligned}
 & \left[?_1 Norm^2 \quad ?_2 Norm^2 \quad ?_3 Norm^2 \quad ?_4 Norm^2 \quad ?_5 Norm^2 \right] * \begin{bmatrix} 1. \text{ SALC in A1} \\ 2. \text{ SALC in A1} \\ 3. \text{ SALC in A1} \\ 4. \text{ SALC in A1} \\ 5. \text{ SALC in A1} \end{bmatrix} = \left[?_1 Norm^2 \quad ?_2 Norm^2 \right] \\
 & = ?_1 Norm^2 \mathbf{s}_1 + ?_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_2}{2} + \frac{\sqrt{2}\mathbf{s}_6}{2} \right) + ?_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_3}{2} + \frac{\sqrt{2}\mathbf{s}_5}{2} \right) + ?_4 Norm^2 \mathbf{s}_4 + ?_5 Norm^2 \mathbf{s}_7 \\
 &)
 \end{aligned}$$

- orbital of A1 with energy = $E_4(A1)$

(

$$[\varphi_1 Norm^2 \quad \varphi_2 Norm^2 \quad \varphi_3 Norm^2 \quad \varphi_4 Norm^2 \quad \varphi_5 Norm^2] * \begin{bmatrix} 1. \text{ SALC in A1} \\ 2. \text{ SALC in A1} \\ 3. \text{ SALC in A1} \\ 4. \text{ SALC in A1} \\ 5. \text{ SALC in A1} \end{bmatrix} = [\varphi_1 Norm^2 \quad \varphi_2 Norm^2$$

$$= \varphi_1 Norm^2 \mathbf{s}_1 + \varphi_2 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_2}{2} + \frac{\sqrt{2}\mathbf{s}_6}{2} \right) + \varphi_3 Norm^2 \left(\frac{\sqrt{2}\mathbf{s}_3}{2} + \frac{\sqrt{2}\mathbf{s}_5}{2} \right) + \varphi_4 Norm^2 \mathbf{s}_4 + \varphi_5 Norm^2 \mathbf{s}_7$$

)

Transitions from p orbitals into s orbitals

State with Occupation [2, 2, 2, 0, 0, 0, 2] (0 unpaired electrons, mult 1.0)

Transition from ψ_2 into ψ_{s_2} :

- $-[2, 2, 2, 0, 0, 0, 2], (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0, 2), (0, 1, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $-p_2 + p_6$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_2 : linear combination = $-s_2 + s_6$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $2E0(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $2E0(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-[2, 2, 2, 0, 0, 0, 2], (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0, 2), (0, 0, 1, 0, 0, 0, 0)$
- ψ_3 : linear combination = $5?_1 Norm^2 p_1$, energy = $E0(B2)$
- η_3 : linear combination = $5?_1 Norm^2 s_1$, energy = $E0(A1)$
- Δ Energy: $E0(A1) - E0(B2)$
- Energy of State before Excitation: $2E0(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $E0(A1) + E0(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

State with Occupation (2, 1, 1, 1, 1, 0, 2) (4 unpaired electrons, mult 5.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 1, 1, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 1, 1, 1, 0, 2), (0, 1, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $-p_2 + p_6$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_2 : linear combination = $-s_2 + s_6$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $E0(B2) + E1(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$

- Energy of State after Excitation: $E0(B2) + E1(B2) + E2(B2) + 2E4(B2) + 2\tilde{\alpha} + \tilde{\alpha}_s + 2\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s3} :

- $-(2, 1, 1, 1, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 0, 1, 1, 0, 2), (0, 0, 1, 0, 0, 0, 0)$
- ψ_3 : linear combination = $5?_1 Norm^2 p_1$, energy = $E0(B2)$
- η_3 : linear combination = $5?_1 Norm^2 s_1$, energy = $E0(A1)$
- Δ Energy: $E0(A1) - E0(B2)$
- Energy of State before Excitation: $E0(B2) + E1(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $E0(A1) + E1(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s4} :

- $-(2, 1, 1, 1, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2), (0, 0, 0, 1, 0, 0, 0)$
- ψ_4 : linear combination = $5?_1 Norm^2 p_1$, energy = $E1(B2)$
- η_4 : linear combination = $5?_1 Norm^2 s_1$, energy = $E1(A1)$
- Δ Energy: $E1(A1) - E1(B2)$
- Energy of State before Excitation: $E0(B2) + E1(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $E0(B2) + E1(A1) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s5} :

- $-(2, 1, 1, 1, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0, 2), (0, 0, 0, 0, 1, 0, 0)$
- ψ_5 : linear combination = $5?_1 Norm^2 p_1$, energy = $E2(B2)$
- η_5 : linear combination = $5?_1 Norm^2 s_1$, energy = $E2(A1)$
- Δ Energy: $E2(A1) - E2(B2)$
- Energy of State before Excitation: $E0(B2) + E1(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$

- Energy of State after Excitation: $E0(B2) + E1(B2) + E2(A1) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 0, 1, 0, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 0, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 0, 1, 0, 2), (0, 1, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $-p_2 + p_6$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_2 : linear combination = $-s_2 + s_6$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $2E0(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $2E0(B2) + E2(B2) + 2E4(B2) + 2\tilde{\alpha} + \tilde{\alpha}_s + 2\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 0, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2), (0, 0, 1, 0, 0, 0, 0)$
- ψ_3 : linear combination = $5?_1 Norm^2 p_1$, energy = $E0(B2)$
- η_3 : linear combination = $5?_1 Norm^2 s_1$, energy = $E0(A1)$
- Δ Energy: $E0(A1) - E0(B2)$
- Energy of State before Excitation: $2E0(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $E0(A1) + E0(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 1, 2, 0, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0, 2), (0, 0, 0, 0, 1, 0, 0)$
- ψ_5 : linear combination = $5?_1 Norm^2 p_1$, energy = $E2(B2)$
- η_5 : linear combination = $5?_1 Norm^2 s_1$, energy = $E2(A1)$

- Δ Energy: $E2(A1) - E2(B2)$
- Energy of State before Excitation: $2E0(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $2E0(B2) + E2(A1) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

State with Occupation (2, 1, 2, 1, 0, 0, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 1, 2, 1, 0, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 0, 2, 1, 0, 0, 2), (0, 1, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $-p_2 + p_6$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_2 : linear combination = $-s_2 + s_6$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $2E0(B2) + E1(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $2E0(B2) + E1(B2) + 2E4(B2) + 2\tilde{\alpha} + \tilde{\alpha}_s + 2\tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 1, 2, 1, 0, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0, 2), (0, 0, 1, 0, 0, 0, 0)$
- ψ_3 : linear combination = $5?_1 Norm^2 p_1$, energy = $E0(B2)$
- η_3 : linear combination = $5?_1 Norm^2 s_1$, energy = $E0(A1)$
- Δ Energy: $E0(A1) - E0(B2)$
- Energy of State before Excitation: $2E0(B2) + E1(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $E0(A1) + E0(B2) + E1(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 1, 2, 1, 0, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 2, 0, 0, 0, 2), (0, 0, 0, 1, 0, 0, 0)$

- ψ_4 : linear combination = $5?_1 Norm^2 \mathbf{p}_1$, energy = $E1(B2)$
- η_4 : linear combination = $5?_1 Norm^2 \mathbf{s}_1$, energy = $E1(A1)$
- Δ Energy: $E1(A1) - E1(B2)$
- Energy of State before Excitation: $2E0(B2) + E1(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy of State after Excitation: $2E0(B2) + E1(A1) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\beta}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 0, 1, 0, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 0, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 0, 1, 0, 2), (0, 1, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $-\mathbf{p}_2 + \mathbf{p}_6$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_2 : linear combination = $-\mathbf{s}_2 + \mathbf{s}_6$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $E0(B2) + E2(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $E0(B2) + E2(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 0, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 0, 1, 0, 2), (0, 0, 1, 0, 0, 0, 0)$
- ψ_3 : linear combination = $5?_1 Norm^2 \mathbf{p}_1$, energy = $E0(B2)$
- η_3 : linear combination = $5?_1 Norm^2 \mathbf{s}_1$, energy = $E0(A1)$
- Δ Energy: $E0(A1) - E0(B2)$
- Energy of State before Excitation: $E0(B2) + E2(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $E0(A1) + E2(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

Transition from ψ_5 into ψ_{s_5} :

- $-(2, 2, 1, 0, 1, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0, 2), (0, 0, 0, 0, 1, 0, 0)$
- ψ_5 : linear combination = $5?_1 Norm^2 p_1$, energy = $E2(B2)$
- η_5 : linear combination = $5?_1 Norm^2 s_1$, energy = $E2(A1)$
- Δ Energy: $E2(A1) - E2(B2)$
- Energy of State before Excitation: $E0(B2) + E2(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $E0(B2) + E2(A1) + 2E4(B2) + 4\tilde{\alpha}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

State with Occupation (2, 2, 1, 1, 0, 0, 2) (2 unpaired electrons, mult 3.0)

Transition from ψ_2 into ψ_{s_2} :

- $-(2, 2, 1, 1, 0, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 1, 1, 1, 0, 0, 2), (0, 1, 0, 0, 0, 0, 0)$
- ψ_2 : linear combination = $-p_2 + p_6$, energy = $\tilde{\alpha} - \tilde{\beta}$
- η_2 : linear combination = $-s_2 + s_6$, energy = $\tilde{\alpha}_s - \tilde{\beta}_s$
- Δ Energy: $-\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy of State before Excitation: $E0(B2) + E1(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $E0(B2) + E1(B2) + 2E4(B2) + 3\tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $\tilde{\delta}$

Transition from ψ_3 into ψ_{s_3} :

- $-(2, 2, 1, 1, 0, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 0, 1, 0, 0, 2), (0, 0, 1, 0, 0, 0, 0)$
- ψ_3 : linear combination = $5?_1 Norm^2 p_1$, energy = $E0(B2)$
- η_3 : linear combination = $5?_1 Norm^2 s_1$, energy = $E0(A1)$
- Δ Energy: $E0(A1) - E0(B2)$
- Energy of State before Excitation: $E0(B2) + E1(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $E0(A1) + E1(B2) + 2E4(B2) + 4\tilde{\alpha}$

- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

Transition from ψ_4 into ψ_{s_4} :

- $-(2, 2, 1, 1, 0, 0, 2), (0, 0, 0, 0, 0, 0, 0) \rightarrow (2, 2, 1, 0, 0, 0, 2), (0, 0, 0, 1, 0, 0, 0)$
- ψ_4 : linear combination = $5?_1 Norm^2 \mathbf{p}_1$, energy = $E1(B2)$
- η_4 : linear combination = $5?_1 Norm^2 \mathbf{s}_1$, energy = $E1(A1)$
- Δ Energy: $E1(A1) - E1(B2)$
- Energy of State before Excitation: $E0(B2) + E1(B2) + 2E4(B2) + 4\tilde{\alpha}$
- Energy of State after Excitation: $E0(B2) + E1(A1) + 2E4(B2) + 4\tilde{\alpha}$
- Energy between Average Energies of ψ -/ η -orbitals: $\frac{E0(A1)}{7} - \frac{E0(B2)}{7} + \frac{E1(A1)}{7} - \frac{E1(B2)}{7} + \frac{E2(A1)}{7} - \frac{E2(B2)}{7} + \frac{E3(A1)}{7} - \frac{E3(B2)}{7} + \frac{E4(A1)}{7} - \frac{E4(B2)}{7} - \frac{2\tilde{\alpha}}{7} + \frac{2\tilde{\alpha}_s}{7}$
- dipole transition moment: $?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta}$

- AND

17. State: A = (2, 2, 1, 1, 0, 0, 2) ↔ B = (2, 2, 1, 0, 1, 0, 2)

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s + 2\tilde{\beta} - 2\tilde{\beta}_s \right)} + \frac{8\tilde{\delta}^2 \left(?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta} \right)}{r^6 \left(E2(A1) - E2(B2) - \tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

18. State: A = (2, 2, 1, 1, 0, 0, 2) ↔ B = (2, 2, 1, 1, 0, 0, 2)

$$E_{dispersion} = \frac{16\tilde{\delta}^4}{r^6 \left(-2\tilde{\alpha} + 2\tilde{\alpha}_s + 2\tilde{\beta} - 2\tilde{\beta}_s \right)} + \frac{16\tilde{\delta}^2 \left(?_1^2 Norm^4 \tilde{\delta} + ?_2^2 Norm^4 \tilde{\delta} + ?_3^2 Norm^4 \tilde{\delta} + ?_4^2 Norm^4 \tilde{\delta} + ?_5^2 Norm^4 \tilde{\delta} \right)}{r^6 \left(E1(A1) - E1(B2) - \tilde{\alpha} + \tilde{\alpha}_s + \tilde{\beta} - \tilde{\beta}_s \right)}$$

Ground-State:

A=[2, 2, 2, 0, 0, 0, 2] ↔ B=[2, 2, 2, 0, 0, 0, 2]

$$E_{dispersion} = -\frac{8\tilde{\delta}^4}{r^6 \left(\tilde{\alpha} - \tilde{\alpha}_s - \tilde{\beta} + \tilde{\beta}_s \right)}$$